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Grupa 341

SSI - Lab 3

I. One Time Pad (OTP)

1. Mesaj Base64:

   o9/khC3Pf3/9CyNCbdzHPy5oorccEawZSFt3mgCicRnihDSM8Obhlp3vviAVuBbiOtCSz6husBWqhfF0Q/8EZ+6iI9KygD3hAfFgnzyv9w==

   Cheie:

   ecb181a479a6121add5b42264db9b44b4b48d7d93c62c56a3c3e1aba64c7517a90ed44f8919484b6ed8acc4670db62c249b9f5bada4ed474c9e4d111308b614788cd4fbdc1e949c1629e12fa5fdbd9

   Rezultate:

   Am transformat mesajul din Base64 in binar, cheia am transformat-o in binar de asemenea, dupa care am aplicat XOR intre ele, dupa care am transformat in text.
   Mai apoi, am luat binarul mesajului criptat, cat si cel al celui decriptat, am facut XOR, am obtinut o cheie, pe care dupa ce am xorat-o la mesajul criptat am obtinut mesajul clar dorit.

   Mesaj in binar:

   10100011 11011111 11100100 10000100 00101101 11001111 01111111 01111111 11111101 00001011 00100011 01000010 01101101 11011100 11000111 00111111 00101110 01101000 10100010 10110111 00011100 00010001 10101100 00011001 01001000 01011011 01110111 10011010 00000000 10100010 01110001 00011001 11100010 10000100 00110100 10001100 11110000 11100110 11100001 10010110 10011101 11101111 10111110 00100000 00010101 10111000 00010110 11100010 00111010 11010000 10010010 11001111 10101000 01101110 10110000 00010101 10101010 10000101 11110001 01110100 01000011 11111111 00000100 01100111 11101110 10100010 00100011 11010010 10110010 10000000 00111101 11100001 00000001 11110001 01100000 10011111 00111100 10101111 11110111

Cheie in binar:

11101100 10110001 10000001 10100100 01111001 10100110 00010010 00011010
11011101 01011011 01000010 00100110 01001101 10111001 10110100 01001011
01001011 01001000 11010111 11011001 00111100 01100010 11000101 01101010
00111100 00111110 00011010 10111010 01100100 11000111 01010001 01111010
10010000 11101101 01000100 11111000 10010001 10010100 10000100 10110110
11101101 10001010 11001100 01000110 01110000 11011011 01100010 11000010
01001001 10111001 11110101 10111010 11011010 01001110 11010100 01110100
11001001 11100100 11010001 00010001 00110000 10001011 01100001 01000111
10001000 11001101 01001111 10111101 11000001 11101001 01001001 11000001
01100010 10011110 00010010 11111010 01011111 11011011 11011001

Rezultatul XOR-at:

01001111 01101110 01100101 00100000 01010100 01101001 01101101 01100101
00100000 01010000 01100001 01100100 00100000 01100101 01110011 01110100
01100101 00100000 01110101 01101110 00100000 01110011 01101001 01110011
01110100 01100101 01101101 00100000 01100100 01100101 00100000 01100011
01110010 01101001 01110000 01110100 01100001 01110010 01100101 00100000
01110000 01100101 01110010 01100110 01100101 01100011 01110100 00100000
01110011 01101001 01100111 01110101 01110010 00100000 01100100 01100001
01100011 01100001 00100000 01100101 01110011 01110100 01100101 00100000
01100110 01101111 01101100 01101111 01110011 01101001 01110100 00100000
01100011 01101111 01110010 01100101 01100011 01110100 00101110

MESAJ DECRIPTAT:

One Time Pad este un sistem de criptare perfect sigur daca este folosit corect.

2. Există o cheie care ar fi decriptat același text criptat de la pct.1 în textul clar următor? Care este această cheie?

   Da, daca folosim un algorirtm OTP, cheia utilizata trebuie sa fie unica si aleatoare, In acest caz, cheia originala este singura care poate realiza aceasta decriptare cu exact acelasi text criptat. Daca dorim o alta cheie, aceasta ar trebui sa genereze acelasi text criptat pentru mesajul original => aplicam XOR intre mesajul decriptat cu Base64 si mesajul dat in binar

   REZULTATUL:

e2bc81f759ae5f1a8e7f466218b2e7524b1bc3dd3c62c97a3a3e03ba66cd106b96e114e59d968ee4e98ed05434

3. Daca 2 texte sunt criptate cu aceeasi cheie si aplicam XOR intre cele 2 texte criptate, cheia se anuleaza si obtinem XOR-ul dintre mesajele inițiale. Asta poate duce la deducerea unei parti din continutul mesajelor, mai ales daca unul dintre mesaje este partial cunoscut. Refolosirea unei chei duce la pierderea caracteristicii principale a unui cifru One Time Pad – aceea ca fiecare bit al mesajului este complet aleatoriu datorita cheii unice, irepetabile.

II. Sisteme de criptare istorice

1. Ex de sistem de criptare prin substitutie -> **CIFRUL LUI CEZAR**
   - CRIPTARE: **ALEXANDRA**, k = 3 => **DOHADQGUD**
   - DECRIPTARE: **VDOXW**, k = -3 (se inverseaza deplasarea) => **SALUT**

   Securitatea sistemului de criptare:
   - Slaba, deoarece sunt doar 25 de chei posibile (in alfabetul latin).
   - Se poate sparge usor prin **forta bruta** sau **analiza frecventiala** (literele cele mai des folosite, cum e E in romana sau engleza, pot da indicii).

2. Ex de sistem de criptare prin transpozitie -> **CIFRUL COLOANELOR**
   - CRIPTARE: **ALEXANDRA** (9 litere)
     - Fie o cheie k = 3 1 4 2 (4 coloane, citim 3->1->4->2)
     - Noi avem 4 coloane -> completam cu _ ca sa avem 12 litere (3 randuri × 4 coloane):
       A L E X A N D R A _ _ _

C1	C2	C3	C4
A	L	E	X
A	N	D	R
A	_	_	_

- Rearanjam coloanele conform cheii 3 1 4 2 (Citim intai coloana 3, coloana 1… -> ED_, AAA, XR_, LN_)
- Citim textul criptat (vertical) -> **EDAAAXRLN_**

- DECRIPTARE: **EDAAAXRLN_** (11 litere), k = 3 1 4 2
  - Adaugam un "_" pentru a avea un tabel de 3 randuri x 4 coloane

Pozitia in criptat	Reprezinta coloana
1 (E D _)	Coloana 3
2 (A A A)	Coloana 1
3 (X R _)	Coloana 4
4 (L N _)	Coloana 2

  - Refacem coloanele si citim randurile:
    Rand 1: **A L E X**
    Rand 2: **A N D R**
    Rand 3: **A _ _ _**

    REZULTAT: **ALEXANDRA**

Securitatea sistemului de criptare:

- Pentru un numar de coloane n, exista **n!** chei posibile. Acest lucru face forta bruta mai dificila. Totusi, pastreaza frecventa literelor si este vulnerabil la atacuri bazate pe permutari.
- Este vulnerabil pentru cuvinte mici pentru ca exista putine chei.
- Tehnici de criptanaliza: **analiza frecventei** si **atacuri bazate pe permutari**

III. Analiza de Frecventa

- Facem frecventa literelor. Cele mai frecvente litere in engleza (aproximativ): **E, T, A, O, I, N, S, H, R, D, L, U**
- Cautam cuvintele cele mai scurte precum "THE", "AND", "OF", "IN", "TO", "IT", "IS", "AS", "AT" etc.

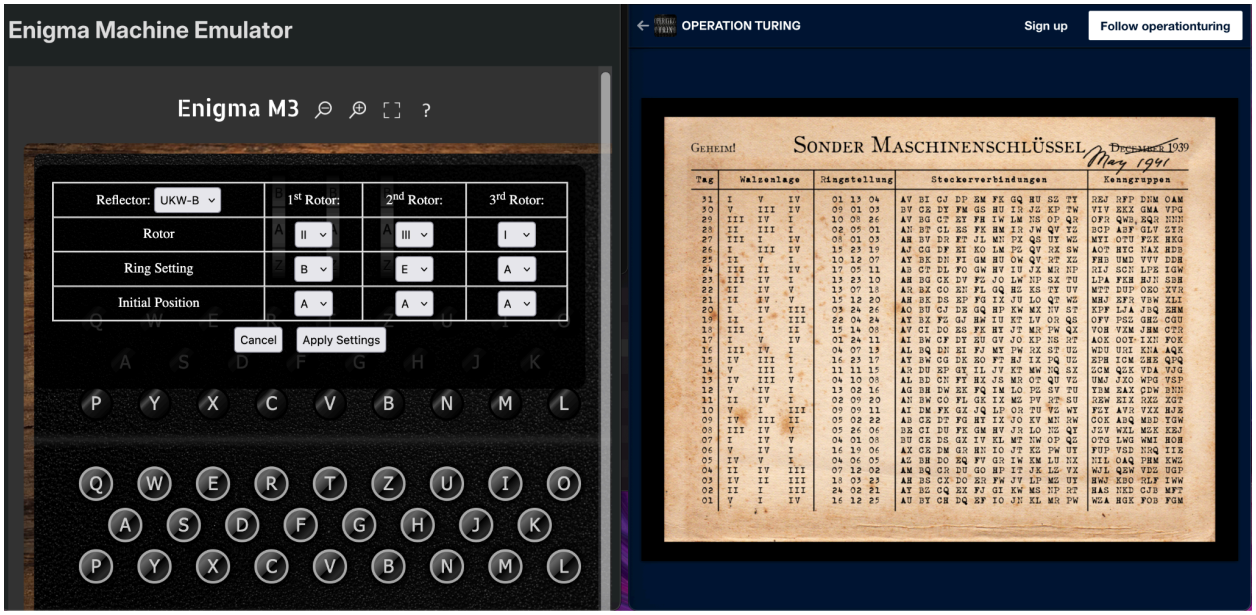
REZULTAT:

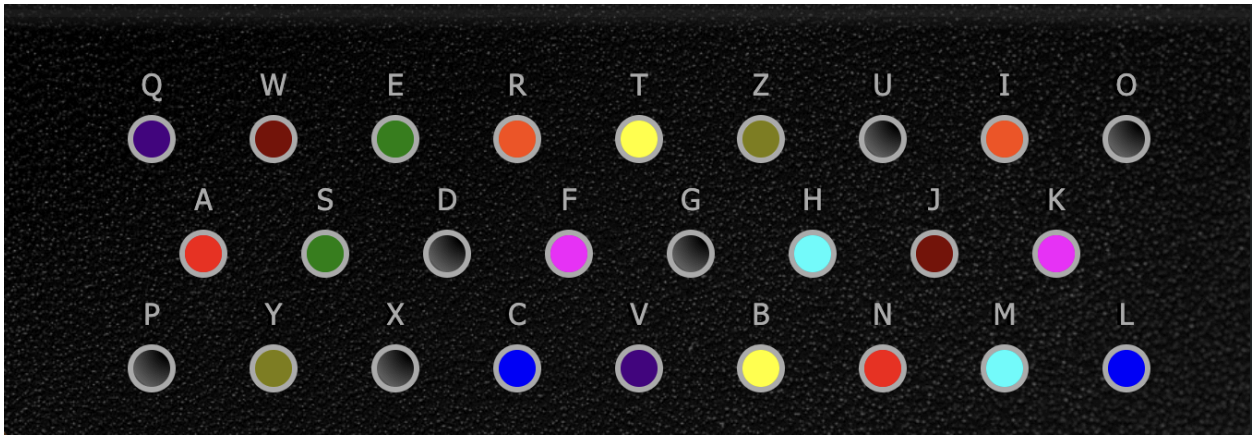
ALICE AND BOB ARE THE WORLDS MOST FAMOUS CRYPTOGRAPHIC COUPLE. SINCE THEIR INVENTION IN 1978, THEY HAVE AT ONCE BEEN CALLED INSEPARABLE, AND HAVE BEEN THE SUBJECT OF NUMEROUS DIVORCES, TRAVELS, AND TORMENTS. IN THE ENSUING YEARS, OTHER CHARACTERS HAVE JOINED THEIR CRYPTOGRAPHIC FAMILY. THERES EVE, THE PASSIVE AND SUBMISSIVE EAVESDROPPER, MALLORY THE MALICIOUS ATTACKER, AND TRENT, TRUSTED BY ALL, JUST TO NAME A FEW. WHILE ALICE, BOB, AND THEIR EXTENDED FAMILY WERE ORIGINALLY USED TO EXPLAIN HOW PUBLIC KEY CRYPTOGRAPHY WORKS, THEY HAVE SINCE BECOME

WIDELY USED ACROSS OTHER SCIENCE AND ENGINEERING DOMAINS. THEIR INFLUENCE CONTINUES TO GROW OUTSIDE OF ACADEMIA AS WELL: ALICE AND BOB ARE NOW A PART OF GEEK LORE, AND SUBJECT TO NARRATIVES AND VISUAL DEPICTIONS THAT COMBINE PEDAGOGY WITH IN-JOKES, OFTEN REFLECTING OF THE SEXIST AND HETERONORMATIVE ENVIRONMENTS IN WHICH THEY WERE BORN AND CONTINUE TO BE USED. MORE THAN JUST THE WORLDS MOST FAMOUS CRYPTOGRAPHIC COUPLE, ALICE AND BOB HAVE BECOME AN ARCHETYPE OF DIGITAL EXCHANGE, AND A LENS THROUGH WHICH TO VIEW BROADER DIGITAL CULTURE. Q.DUPONT AND A.CATTAPAN CRYPTOCOUPLE

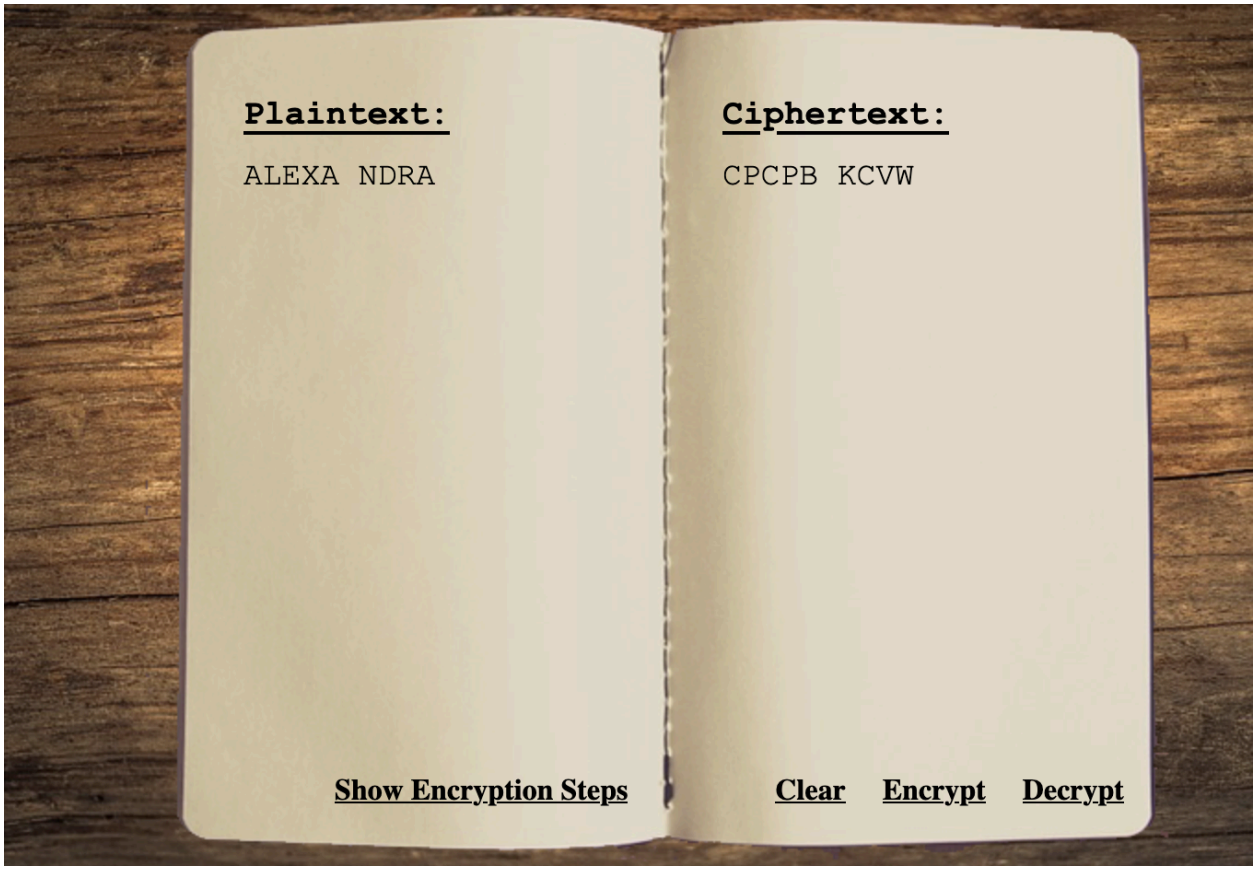
IV. Sistemul de criptare mecanic Enigma

1, 2 - **data zilei: 28.03**

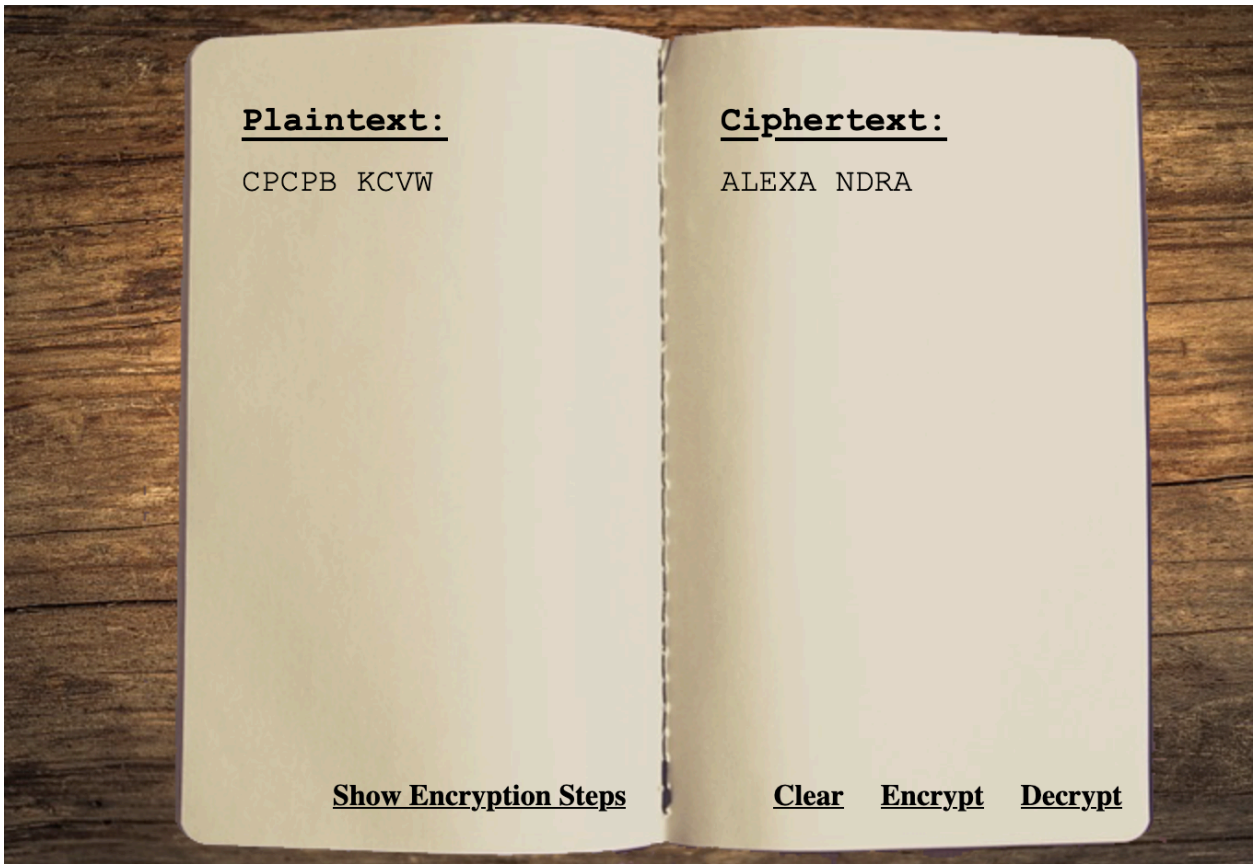




3.


Plaintext:
ALEXA NDRA
Ciphertext:
CPCPB KCVW
Show Encryption Steps
Clear
Encrypt
Decrypt

4.



- Se da **Clear**, dupa care se alege optiunea de **Decrypt**. Inainte de decritare, trebuie sa ne asiguram ca **pozitia initiala** la roti este setata la fel ca la criptare.

5. Fie **ALEXANDRA**, textul deja mai sus criptat (9 litere). Luam un text aleator de aceeasi lungime cu ALEXANDRA (ex: **BBBBBBBBB**) si il criptam cu aceleasi setari:



Observam ca la criptare, chiar daca am avut un string de aceeasi lungime, masina a dat o criptare total diferit => Motivul pentru care se intampla acest lucru este din cauza reflectorului care atunci cand apas aceeasi tasta de mai multe ori, prin redirectionarea curentului electric printr-un traseu diferit, acesta nu va permite niciodata sa reflecte litera “B” in aceeasi litera.